
Alejandro Pérez Velilla

Curriculum Vitae

Positions

2026– **Postdoctoral Research Fellow**, *National University of Singapore*, Singapore
Using cultural evolutionary tools to develop cooperative, culturally-aligned AI systems.

Education

- 2021–2026 **PhD studies in Cognitive and Information Sciences**, *University of California, Merced*, Merced, California, USA
Ongoing PhD studies under the supervision of Paul Smaldino at University of California, Merced.
- 2015–2020 **Licenciatura in Anthropological Sciences, Sociocultural Orientation**, *Universidad de Buenos Aires*, Buenos Aires, Argentina
Licentiate degree program in Anthropological Sciences. Five-year program ending with a graduate thesis defense.
- 2014–2015 **Ciclo Básico Común**, *Universidad de Buenos Aires*, Buenos Aires, Argentina
Basic common cycle, necessary for admission to a Universidad de Buenos Aires degree program.
- 2011–2012 **Filmmaking Year Program**, *Prague Film School*, Prague, Czech Republic
Intensive one-year program in filmmaking, with emphasis in Direction and Direction of Photography.

Licentiate thesis

- title *Self-organization strategies in scout movements: a case study of the Exploradores movements from the Buenos Aires villas.*
- supervisors Diego Diaz Cordova, PhD.
- description Thesis resulting from a six-month fieldwork with two scoutist Exploradores groups from at-risk neighborhoods (villas) of Buenos Aires. The thesis seeks to provide a possible explanation for the rapid and expansive propagation of scoutism since its introduction at the beginning of the twentieth century. A broad comparison is done between scoutist movements and other prestige hierarchies in the ethnographic literature. It was presented and defended in the first quarter of 2020.

PhD thesis

- title *Cultural evolution under uncertainty and risk.*
- supervisors Paul E. Smaldino, PhD.

description Thesis exploring the evolutionary underpinnings of the human capacity for extensive use of social information under conditions of risk and uncertainty and the downstream consequences for the stability of cooperative arrangements. The thesis explores these themes with a combination of mathematical models, ABMs and simulation-based inference methods for model fitting and inference with ethnographic data.

Publications

In preparation for submission

Pérez Velilla, A., Padilla, K. J., Cassiani Cervantes, D., Herrera, Y., Hernández, A., Hurtado Manyoma, A., Pérez Salinas, M., Sosa, S., Dalla Ragione, A., Ross, C. T. *A full-population assessment of Palenquero proficiency in San Basilio de Palenque, Colombia.*

Pérez Velilla, A., Nöldeke, G., Peña, J. *Reciprocity and the evolution of risk-reduction sharing.*

Preprints

- 2026 Pérez Velilla, A., Ready, E. *The emergence of sharing networks through indirect signaling.* https://osf.io/preprints/socarxiv/6zfju_v1

In press

- 2026 Pérez Velilla, A., Smaldino, P. *Risk, learning and culture.* In A. Gallup, S. Karithey, G. Geher. (Eds.) *Oxford Handbook of Evolutionary Psychology.* Oxford University Press.

Published work

- 2026 Pérez Velilla, A., Smaldino, P. *A demographic theory of similarity-biased learning.* PNAS. <https://www.pnas.org/doi/10.1073/pnas.2606741123>
- 2026 Pérez Velilla, A., Smaldino, P. *The adaptive role of peer culture is shaped by risk landscapes.* Commentary for target article Lew-Levy, S. & Amir, D. (2024) *Children as agents of cultural adaptation.* Behavioral and Brain Sciences. Cambridge University Press.
- 2026 Pérez Velilla, A., Beheim, B., Smaldino, P. *The development of risk attitudes and their cultural transmission.* Psychological Review. American Psychological Association. <https://doi.org/10.1037/rev0000599>
- 2025 Smaldino, P., Pérez Velilla, A. *The Evolution of Similarity-Biased Social Learning.* Evolutionary Human Sciences. Cambridge University Press. 2025;7:e4. doi:10.1017/ehs.2024.46
- 2024 Smaldino, P., Moser, C., Pérez Velilla, A., Werling, M. (2024) *Maintaining transient diversity is a general principle for improving collective problem solving.* Perspectives on Psychological Science 19(2), 454-464.
- 2022 Pérez Velilla, A., Moser, C., Smaldino, P. (2022) *Hidden Clusters Beyond Ethnic Boundaries.* Commentary for target article Uchiyama, Spicer & Muthukrishna (2022) *Cultural Evolution of Genetic Heritability.* Behavioral and Brain Sciences. Cambridge University Press.

- 2020 Pérez Velilla, A. *Redes sociales y jerarquización: dinámicas de relación y diferenciación en un movimiento escautista bonaerense*. Presentation and book chapter for Reunión Latinoamericana de Redes Sociales. Mendoza, Universidad Nacional de Cuyo.
- 2019 Pérez Velilla, A. *Jerarquía y desigualdad social: reflexiones antropológicas en torno al rol de las estructuras sociales jerárquicas en la organización de la vida social*. Presentation for Segundas Jornadas de Estudio Sobre Jerarquías. Buenos Aires, Facultad Latinoamericana de Ciencias Sociales.
- 2018 Pérez Velilla, A. *Dinámicas de auto-organización en un movimiento escautista bonaerense*. Presentation for Work Group 17: Anthropological studies on social movements: practices, politics and relationships with state agencies. IX Jornadas de Investigación Santiago Wallace. Facultad de Filosofía y Letras, Universidad de Buenos Aires.
<http://jiassw.com.ar/wp-content/uploads/2017/01/IX-JIASSW-Programa.pdf>

Grants and awards

- 2024 Graduate Student Researcher Fellowship. *Environmental Instability and the Utility of Prestige-Biased v. Conformist Social Learning*. Center for Analytic Political Engagement.
- 2024 Advancing Cultural Evolution (ACE) Early Career Research Grant. *The formation of economic aid networks and their effects on language revitalization efforts in San Basilio de Palenque, Colombia*. Cultural Evolution Society.

Academic experience

Participation in academic conferences and workshops

- 2025 **Building a Science of Cultural Evolution for the 21st Century Workshop**, *Santa Fe Institute*, Santa Fe, New Mexico, USA
 Co-organizer
- 2025 **California Workshop on Evolutionary Social Sciences**, *University of California, Santa Barbara*, Santa Barbara, California, USA
 Presentation: The development of risk behaviors and their cultural transmission
- 2023 **Early Career Social Learning Researchers Society Workshop**, *Max Planck Institute for Human Development*, Berlin, Germany
 Poster: Deceived by Chance: Pessimism in Learning and the Adaptive Value of Conformity Under Uncertainty
- 2023 **Workshop: Information Architectures as Socio-Technical Competitions**, *Santa Fe Institute*, Santa Fe, New Mexico, USA
 Presentation: Social learning, decision-making under uncertainty and environmental instability
- 2023 **European Human Behaviour and Evolution Association conference**, London, United Kingdom
 Presentation: Within-lineage marriage and wealth inheritance
- 2022 **Cultural Evolution Society Conference**, Aarhus, Denmark
 Presentation: The evolution of sharing networks under risk and uncertainty

- 2022 **International Conference of Computational Social Sciences**, Chicago, Illinois
Poster presentation: The evolution of sharing networks under risk and uncertainty
- 2022 **Summer School Complexity-GAINS on Disintegration of Societies**, Vienna, Austria, Santa Fe Institute
- 2022 **Toulouse Summer School in Quantitative Social Sciences**, Toulouse, France, Toulouse School of Economics
- 2022 **Spring School on Evolution of Social Complexity**, Vienna, Austria, Complexity Science Hub Vienna
- 2022 **9th Toulouse Economics and Biology Workshop**, Toulouse, France, Toulouse School of Economics
Poster presentation: The evolution of sharing networks under risk and uncertainty.
- 2019 **Reunión Latinoamericana de Análisis de Redes Sociales**, Mendoza, Argentina, UNCuyo
Latin-American meeting for the use and application of Network Analysis in social research. Presented a network analysis of scout group structure and coordination based on advances in control theory.
- 2019 **Segundas Jornadas de Estudio Sobre Jerarquías**, Buenos Aires, Argentina, FLACSO
Conference on social hierarchies and social inequality. Presented a review of hierarchies as studied in social sciences, particularly anthropology.
- 2019 **Jornadas de Fundamentos y Aplicaciones de la Interdisciplina**, Buenos Aires, Argentina, Universidad de Buenos Aires
Conference on interdisciplinary research. Did a presentation on Complex Systems for the integration of social and biological sciences.
- 2018 **IX Jornadas de Investigación Santiago Wallace**, Buenos Aires, Argentina, Universidad de Buenos Aires
Anthropology congress at University of Buenos Aires. Presented an advance of the study of group processes through a cognitive ethnographic perspective.
- [Positions and internships](#)
- 2024–2026 **Mentorship with undergraduate students from the Cognitive and Information Sciences department at University of California, Merced.**, *Mentorship for Under-Graduates (MUG) Program.*, Merced, California.
- 2024–2026 **Fieldwork in the municipality of San Basilio de Palenque. Demographic, socio-economic and language proficiency census.**, *Cody Ross, Max Planck Institute for Evolutionary Anthropology.*, Bolívar, Colombia.
- 2023 **Fieldwork in the Canadian Arctic Inuit community of Kangiqsujuaq. Household survey on hunting, sharing and food security.**, *Elspeth Ready, Max Planck Institute for Evolutionary Anthropology.*, Quebec, Canada.
- 2022–2023 **Research Assistant for project on modeling Information Architectures**, *Paul Smaldino, University of California Merced.*, Merced, California.
- 2021–
Ongoing **Teacher's Assistant for undergraduate courses from the Cognitive and Information Sciences department**, *Paul Smaldino, University of California Merced.*, Merced, California,

- 2021– Ongoing **Cognitive and Information Sciences PhD program**, *Paul Smaldino, University of California Merced.*, Merced, California.
- 2018–2020 **Distributed fieldwork in two Buenos Aires villas (emergent at-risk neighborhoods).**, Buenos Aires, Argentina.
- 2018 **Teacher’s Assistant for extension seminar “Computational Social Science for the study of Cultural Evolution”**, *Gustavo Landfried, UBA*, Buenos Aires, Argentina.
- 2016–2017 **Internship at Biological Anthropology Laboratory, Universidad Maimónides, Cristina Dejean, Dr. Sergio Avena**, Buenos Aires, Argentina.
DNA extraction and sequencing for unveiling African heritage in the Buenos Aires Metropolitan Area.
- 2015 **Internship at Argentine Team of Forensic Anthropology**, *EAAF*, Buenos Aires, Argentina
- Scientific software development
- 2024 **BaryPlots.jl**, *Julia package for visualizing trajectories of evolutionary games on a ternary simplex.*, Alejandro Pérez Velilla and Jorge Peña., <https://github.com/datadreamscorp/BaryPlots.jl>

Academic interests

- Mathematical and computational modeling of social/cultural systems, institutions and cooperative arrangements and their evolution
- Short- and long-term social norm change under risk and uncertainty
- Evolutionary game theory, numerical simulation and Agent-Based Modeling
- Bayesian statistical modeling and causal inference
- Simulation-based inference for real-world application of computational models

Computer skills

Basic	Spreadsheets, Git, LaTeX, Linux
Modeling and data analysis	Julia, Python, R, SQL
Programming languages	Julia, C++, Java, R, Python, Haskell

Languages

Spanish	Native language	
English	Second language	<i>Band 8.5 IELTS Academic exam, 2018.</i>
Portuguese	Basic	<i>Reading, writing and basic conversational.</i>
German	Basic	<i>Reading, writing and basic conversational.</i>
Mandarin Chinese	Basic	<i>Reading, writing and basic conversational.</i>

Personal interests (hobbies)

Philosophy and Mathematics
Film and Filmmaking
Game design and player data analytics

References

Academic

- Paul Smaldino, PhD., University of California, Merced,
psmaldino@ucmerced.edu
- Diego Díaz Cordova, PhD., Universidad de Buenos Aires, Universidad Nacional de Lanús,
didibart@gmail.com

Internships

- Patricia Bernardi, Argentine Team of Forensic Anthropology,
patobernardi@yahoo.com
- Cristina Dejean, Biological Anthropology Lab, Universidad Maimónides,
dejeancr@gmail.com